

AutoSAM for Video Salient Object Detection

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Abstract

Video Salient Object Detection (VSOD) is an advanced segmentation task that is focused on segmenting the most important objects in videos. VSOD is a key task in the field of computer vision, with many potential applications in a variety of fields. Despite that, existing models often struggle with dynamic scenes, rapid changes, and the inherent subjectivity of saliency. Inspired by Autosam [7], which enhances the Segment Anything Model (SAM), we propose the AutoSAM2-VSOD model, which adapts the recent SAM2 network to the VSOD tasks, making minimal changes to the model in order to keep the high accuracy of SAM2 [5]. Our approach builds upon SAM2 and incorporates video depth and RGB features as well as temporal data in order to improve performance in VSOD tasks, achieving competitive results on the ViDSOD-100 dataset. Our work highlights the potential of adapting general segmentation models to VSOD tasks and demonstrates areas for future refinement to further improve accuracy and applicability.

1 Introduction

Video Salient Object Detection (VSOD) is the task of detecting and tracking the most important object in videos- including understanding context and temporal information. It has varied applications in many fields, including video editing, autonomous driving, medical imaging, ecological research,

and other areas which require the detection of subjectively important objects. Since VSOD requires identifying and segmenting salient objects (while also maintaining temporal consistency), it is closely related to classic segmentation tasks. However, traditional segmentation models, while sufficient for static images, cannot capture temporal changes or object continuity when used directly on VSOD tasks, limiting their applications in that area.

AutoSAM has proven that static image segmentation models such as SAM (segment anything model) can be adapted toward the out-of-domain medical imaging field by incorporating adaptive prompt tuning [7].

Inspired by this approach, we proposed a similar solution, adapting the recent SAM2 model to the VSOD task. We use SAM2 instead of SAM, due to its temporal processing, enhanced feature extraction, and better generalized performance across many datasets [5]. In order to adapt SAM2 to VSOD, we have modified SAM2 by integrating optical flow and depth data to improve its capabilities to detect and track the salient object in videos. Most of the SAM2 model is kept intact, with only the prompt encoding modified, in order to benefit from SAM2’s existing training.

Our work aims to create a model which can compete with the results of other state-of-the-art models while only requiring minimal training due to inheriting SAM2’s capabilities. This will serve to demonstrate the potential flexibility of the SAM2 mode and further show the potential of adapting

segmentation models toward VSOD tasks. Finally, we explore potential directions for improving the model and the potential for better integration of different inputs beyond RGB and depth.

2 Related Work

The Segment Anything Model (SAM) [3] is a powerful vision model developed by Meta AI for universal image segmentation. It segments selected objects in an image indicated by user input, including points, boxes, or masks as prompts. Two of the enhancements to SAM [3] that are relevant to the VSOD task are the SAM2 [5] model and the autoSAM [7] model, with each model enhancing SAM in a different direction.

SAM2 is an extension of the SAM designed to enhance segmentation accuracy and efficiency and is particularly useful in video segmentation and dynamic environments where objects change over time. However, the models’ reliance on user input makes it insufficient for the VSOD task.

AutoSAM is an extension of SAM that eliminates the need for manual prompts by introducing an image-based encoder for automatic segmentation. Originally designed to improve medical image segmentation, AutoSAM enables precise object identification without user interaction - By replacing SAM’s manual conditioning mechanism with automated prompt generation, AutoSAM enhances efficiency, adaptability, and usability in dynamic environments. It achieves state-of-the-art segmentation without requiring fine-tuning, making it effective for tasks where zero-shot learning and automation are crucial. However, autoSAM is adapted to medical images segmentation instead of videos [7] - therefore it is better to use it as a template for adapting SAM2 (which can handle video tasks) toward automatic saliency tasks.

Many works have shown the importance of multi-modal inputs for improving VSOD solutions. One such example is RealFlow [1], a recent model, that achieves state-of-the-art results on the prominent DAVSOD test set by training on generated image-optical flow pairs. Another notable contribution is the ViDSOD-100 dataset [4], which was created to enable training networks on labelled VSOD scenes with ground-truth depth maps. Results on this dataset demonstrated benefits from incorporating depth information into VSOD solutions.

We follow the example of AutoSAM, while adapting SAM2 instead of SAM, with our key changes being an auto prompt generator for saliency, and integrating depth into the prompt generator, which is missing from the SAM2 inputs. We aimed to keep most parts of SAM2 unchanged, training only the prompt generator weights on the VSOD task.

3 Data

Our model is trained on the recent ViDSOD-100 [4] dataset. The dataset consists of 100 scenes, 60 in the training set and 40 in the test set. Each scene contains 20 – 100 frames, and each frame has a respective RGB image, depth map, and saliency mask. The data contains a wide variety of indoor scenes.

Each scene contains 1 or more salient objects, including animals, people, and everyday objects. Saliency appears to be affected by factors such as object type, location in the image, size, distance, and movement speed, among others.

To add optical flow data, we use the optical flow estimates included in the ViDSOD-100 dataset.

An example of the samples from ViDSOD-100 dataset can be seen in Fig 1.

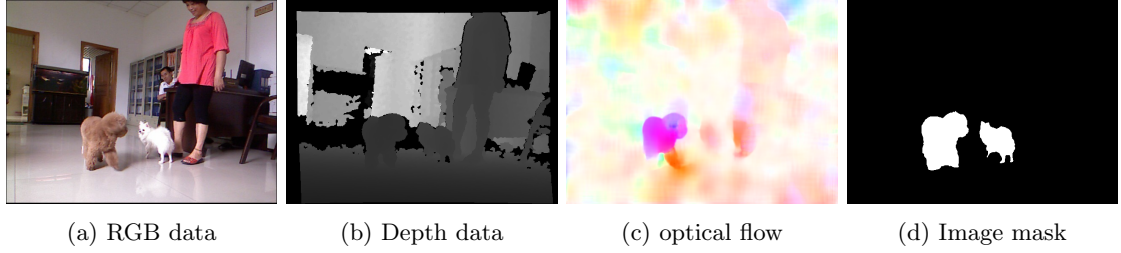


Figure 1: sample frame from ViDSOD-100 dataset.

In earlier experiments, we trained and tested our model independently on the DAVSOD dataset [2]. This dataset is a prominent benchmark for VSOD, with 100 training and scenes and 80 test scenes. Unlike DAVSOD, the dataset does not contain ground truth depth maps.

4 Methods

As described above, our work focuses on replacing the SAM2 user-based prompt encoder [5] with an automatic prompt generator. For the prompt generator, the main challenge is assumed to be identifying the salient objects in the image and encoding this information as a generated prompt, without needing to generate a full segmentation mask at pixel-level accuracy.

We did not update the weights of the core SAM2 model during training, due to the already powerful capabilities of the model, and its state-of-the-art segmentation capabilities. In addition, the SAM2 Vision Transformer (ViT) is a large model, and expensive to train. Instead, like in AutoSAM, our contribution is confined to generated prompt encoding sent to the mask decoder, replacing the encoded user prompt.

Our work uses the 'large' ViT backbone for SAM2 [5], the largest and most effective of the 4 available backbones, to maximize the quality of the output saliency mask.

Past works have used optical flow or other

temporal data for the VSOD task. Since SAM2 already processes temporal data, using cross-attention between the current and previous frames, we attempted to train the prompt generator without explicit temporal data. Following previous works showing the importance of depth for VSOD [4], we incorporated depth into encoder inputs to assist in identifying salient objects.

We use the ESANet [6] as a simple RGB-D encoder, with separate ResNet-34 backbones for encoding RGB and depth information independently. The extracted features from both paths are fused at each resolution, enhancing depth-aware segmentation, and passed through skip connections to the prompt generator decoder to retain spatial details.

When using only RGB and optical flow inputs, we used independent HarDNet backbones for each modality, instead of ESANet.

A variant of the model follows AutoSAM in using the Harmonic Dense Net (HarDNet) for the RGB input in an independent encoder, while using ESANet to fuse the optical flow and depth data. The outputs from the two decoders are concatenated, followed by a channel reduction via a 1×1 convolution before the decoding stage.

A schematic of our solution is included in Fig 2. A more detailed illustration of the architecture for the prompt generator, when using all 3 input modalities, is included in Fig 3.

To normalize the depth maps, we calculated

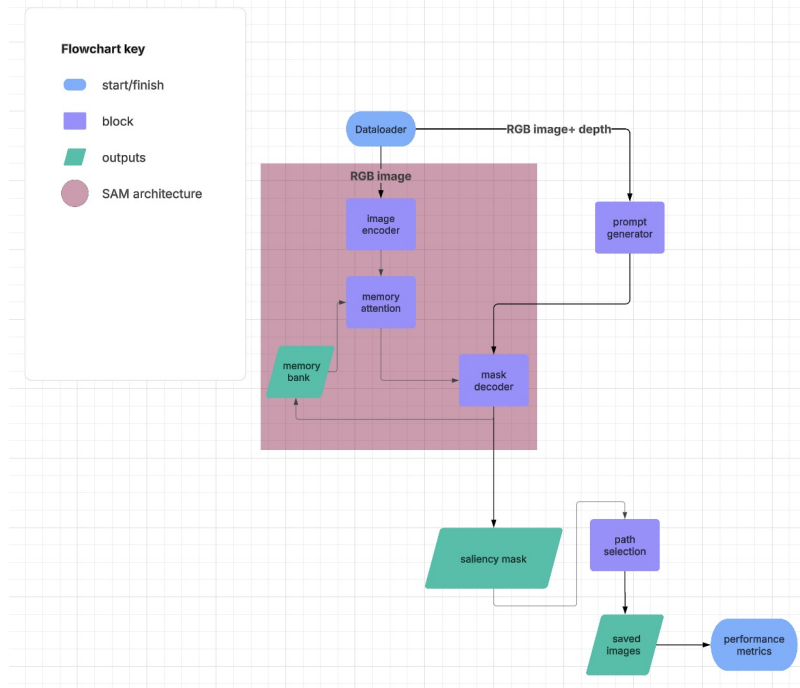


Figure 2: An illustration of AutoSAM2. We replace SAM2 (red, left) with the prompt generator (right). The remaining architecture, comprised of the Image Encoder, Memory Bank, Memory Attention and Mask Decoder, is left unchanged. Only the prompt generator weights are updated during training.

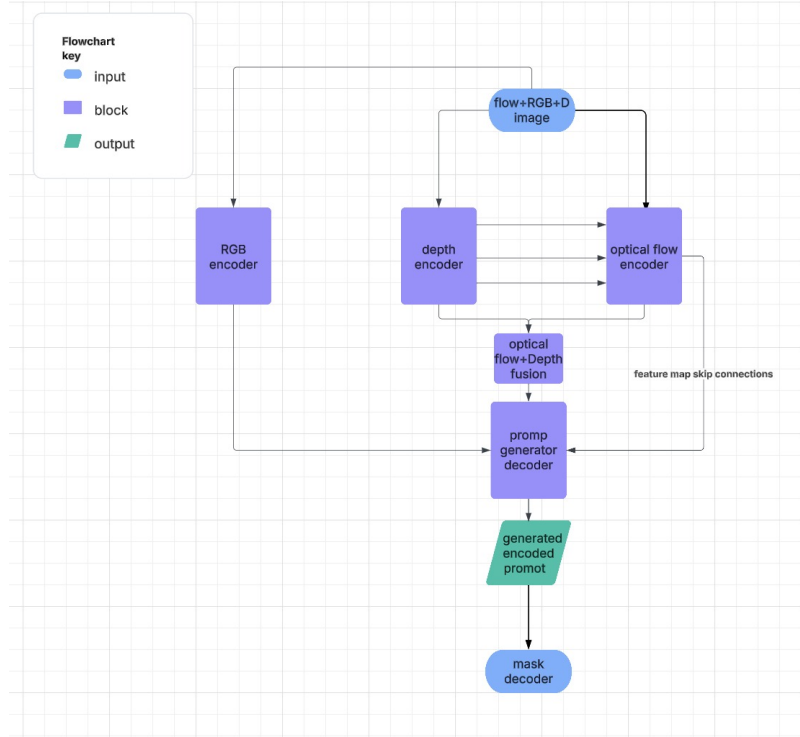


Figure 3: An illustration of the prompt generator architecture. The RGB encoder (left) independently process the RGB input, while the ESANet (right), jointly processes the depth and optical flow inputs. The multi-resolution outputs of the two encoders are concatenated and sent to the prompt generator decoder, which in turn calculates the generated prompt for SAM2.

the mean and variance across the training data, and normalized the depth to a mean of 0 and standard deviation of 1. The optical flow data is normalized with the standard ResNet normalization for RGB images, following the method used for the ATF-Net in ViDSOD-100 [4].

To avoid the complexity of adding temporal information to the prompt encoder, we confined the temporal processing to the SAM2 ViT backbone, with the prompt generator processing each frame independently, aside from the optical flow input. This partially follows the architecture of the ATF-Net [4], which achieves state of the art results with RGB [4], depth and optical flow inputs, without explicit use of additional temporal processing.

To avoid overfitting on the data, we used a number of augmentation, including color jitter, noise, and affine transformations. Following the approach outlined in [4], we train the model exclusively on the ViDSOD-100 dataset, which contains 60 training scenes with a wide variety of content.

For each training or inference scene, we first run the prompt generator independently on each input frame, creating a prompt for each time step. Next, we run the SAM2 ViT to process each frame independently. Finally, we send both the memory attention layer output and the encoded prompt to the pre-trained SAM2 mask decoder, giving us the output logits for the saliency mask.

The resulting output mask is normalized using min-max normalization:

$$X_{\text{norm}}(X) = \frac{X - X_{\min}}{X_{\max} - X_{\min}}. \quad (1)$$

We then compare the normalized output to the ground truth saliency mask using two loss functions. The first is a standard binary cross entropy loss. The second is the

negative of the F-measure score:

$$F_{\beta} = (1 + \beta^2) \cdot \frac{\text{precision} \cdot \text{recall}}{\beta^2 \cdot \text{precision} + \text{recall}}. \quad (2)$$

with $\beta = 0.3$, giving a higher weight to precision.

5 Experiments

Due to the expense and time needed to run models on the full test set consisting of thousands of frames, only the validation loss was used for evaluating most experiments, with the full test set run for a limited number of trained models.

To reduce training time, we edited SAM2 to save gradient for the backwards pass only for the mask decoder, discarding gradients for the ViT, memory attention, and other sections.

Training details We use a learning rate of $3\text{e-}4$ with a decay rate of $1/3$ every 20 epochs, which was found to minimize divergence of the validation loss over time. We find that the validation loss still begins to diverge or oscillate around 30-60 epochs, so we train the model for 100 epochs and take the trained model with the lowest validation loss.

We began by adding a standard sigmoid function on the output of the model. However, since the SAM2 model is pretrained, the model does not converge with this setup. This seems to be due to SAM2 being weighted towards negative values for all outputs for an untrained prompt generator. Instead, following AutoSAM, the normalization defined in (1) is used for each output.

An additional requirement found was clamping the output logits. We add clamping to the range of $(-32, 32)$ to make the above normalization effective.

To potentially reduce the training and inference time of the model, we also trained

Model	Maximum F-measure
ATF-Net1 – Small	0.787
ATF-Net1 – Large	0.864
AutoSAM2 - depth	0.720
AutoSAM2 - depth, frozen encoder	0.7498
AutoSAM2 - depth and optical flow	0.777
AutoSAM2 - optical flow	0.798

Table 1: Performance comparison of AutoSAM2-VSOD and ATF-Net on the ViDSOD-100 test set.

our solution with the smaller base+ backbone to see if the run time could be reduced without significantly reducing the output quality. This was unsuccessful; there was a significant reduction in the validation score compared to the large ViT backbone.

The inputs for each dataset were padded to a square 640×640 image, to match the proportions of the both the generated prompt and SAM2 input [5], avoiding distortion of the image. For SAM2 we further upsampled the image to the default input size of 1024. For the prompt generator, we tested sizes of 256 and 512, with 512 giving the best results. The full 640×640 image is not significantly downsampled, and removing downscaling altogether would have required significantly reducing the batch size during training.

verged, in spite of the consistent decline of the train loss. This indicated issues with overfitting in the model. The set of image augmentations from AutoSAM [7], as well as the natural variation between tens of frames per training scene, were insufficient to solve this issue. To address this, we use Gaussian and salt-and-pepper noise to augment the training images, as well as a lower learning rate, and weight decay.

Some of our initial experiments were run using the DAVSOD dataset for training and testing. However, as we used only an RGB input, without estimated optical flow or depth, we were not able to match the state-of-the-art results on the test set, as expected. We then moved to the ViDSOD-100 test set, which has both ground truth depth maps, and high quality estimates for optical flow.

Our intention had been to use only depth and RGB inputs for the prompt generator, relying on SAM2 for temporal processing. However, the trained model failed to reach state-of-the-art results on the ViDSOD-100 dataset [4]. As a result, we trained 3 variations of our model:

- RGB and depth
- RGB and optical flow
- RGB, depth and optical flow.

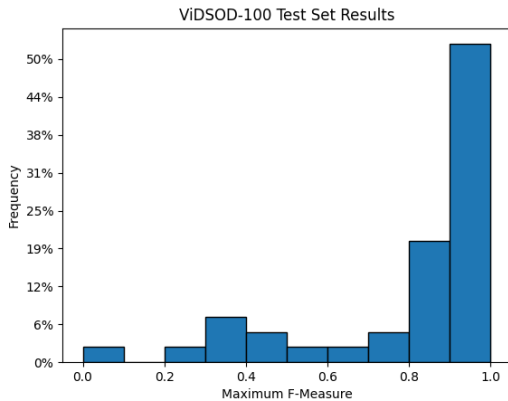
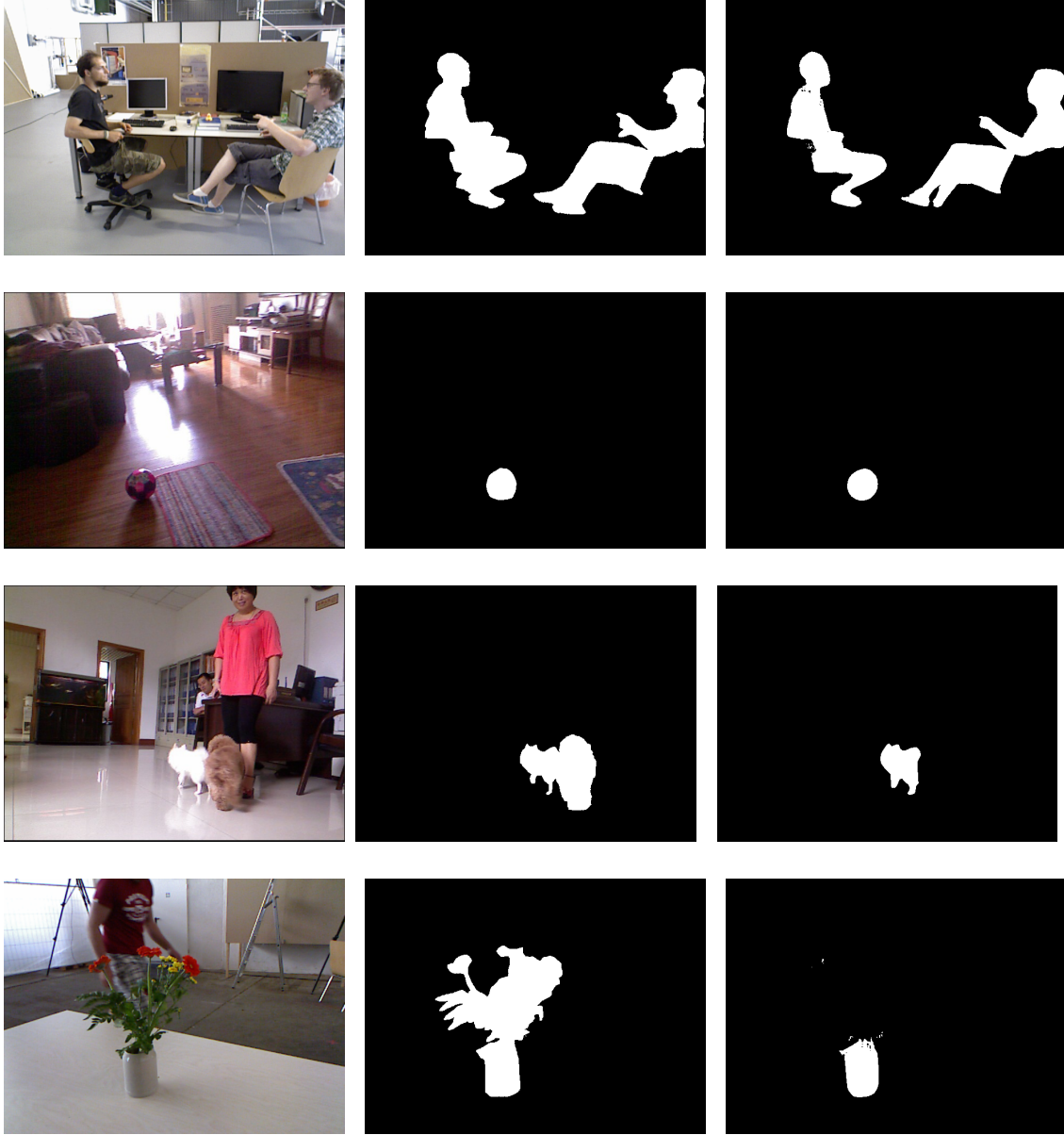


Figure 4: Histogram of AutoSAM2 results by scene.

A frequent problem in training the model was that the validation loss quickly di-

The addition of optical flow significantly improved the F-measure score, highlighting the importance of object movement to video saliency.



(a) RGB data

(b) Image mask

(c) AutoSAM2's result

Figure 5: Sample AutoSAM2 results on ViDSOD-100 frames.

For the variant with optical flow, we tested variants with both ESANet and independent processing with 2 HardNet encoders. Of these, using 2 HardNet encoders gave better results.

For the variant with all 3 input modalities, we tested a number of architectures, including both adding an independent optical flow encoder to in parallel to the ESANet encoder, and processing all 3 modalities independently. Empirically, the best results came from processing the optical flow and depth with ESA-Net, and processing the RGB input independently with HardNet.

Evaluation metrics We use the F-measure for evaluating our model, defined in (2), Giving in this case a higher weight towards precision vs. recall. We also considered using the Mean Average Error (MAE). However, since we use a linear min-max normalization of the output, instead of a Sigmoid, the metric was not shown to be a meaningful evaluation of the output.

Results The model was evaluated on the test set of the ViDSOD-100 dataset. We compared our results to the current state-of-the-art for the ViDSOD-100, ATF-Net. The results are shown in Table 1. Results for both the small and large variants of ATF-Net are included

Our best results come from combining only the RGB and optical flow, with similar results from adding depth. This indicates that our model is not making effective use of the depth input. Our highest score is greater than that of the small ATF-Net, but notably outperformed by the more powerful large ATF-Net.

In addition, we include a histogram of the Maximum F-measure score for each scene in the test set in Fig 4. Our model has a score in the 90th percentile for just over 50% of the scenes, and has a score below 0.5 in under 20% of scenes.

Visual samples of our results on the

ViDSOD-100 dataset are included in Fig. 5.

We included samples that demonstrate both the strengths of the trained model, and areas where it faces challenges. In first top scene, the model successfully segments both salient persons with a high degree of precision. In the second scene, it recognizes the salient ball despite a busy indoor scene.

In the third scene, the model correctly ignores both non salient individuals, but segments only the white dog, failing to segment the brown dog on the left. Finally, in the fourth scene, the model struggles to correctly segment the entire flower in the foreground.

6 Conclusions

In this work, we proposed an adaptation of SAM2 to the Video Saliency (VSOD) task, leveraging the powerful and versatile SAM2 model combined with a simple automatic prompt generator to train a competitive VSOD solution. By keeping most of the original SAM2 architecture intact, and avoiding finetuning the ViT backbone, our solution offers a relatively lightweight training process while maintaining SAM2’s accurate segmentation capabilities.

Our results on the ViDSOD-100 dataset demonstrate the potential of this approach. They also demonstrate the importance of temporal processing for identification of salient objects in video, with a significant increase in accuracy when adding this data to the prompt generator inputs. However, despite the strong performance of our solution, we did not achieve state of the art results on the ViDSOD-100 dataset, showing the need for further improvements. These challenges, despite the SAM2 model’s state of the art results for the Segment Anything task, highlights the unique challenges of VSOD.

Further work may explore better integration

of the different input modalities, potentially both reducing the model size and enhancing performance. Nonetheless, we hope that the results achieved with our approach will serve as a foundation for further expanding SAM2’s capabilities in VSOD and other novel fields with efficient training processes.

References

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- [7] Tal Shaharabany, Aviad Dahan, Raja Giryes, and Lior Wolf. Autosam: Adapting SAM to medical images by overloading the prompt encoder. *CoRR*, abs/2306.06370, 2023.

7 appendix

The code for our work can be found at <https://github.com/yedkap/AutoSAM2-VSOD-personal>.

We started our project using the code from the AutoSAM project github page. We also worked with the code for SAM and SAM2, and ESANet. Noise augmentations were based on those from the ViDSOD-100 code.

Links to the above are included below:
<https://github.com/talshaharabany/AutoSAM>

<https://github.com/facebookresearch/segment-anything>

<https://github.com/facebookresearch/sam2>

<https://github.com/TUI-NICR/ESANet>

<https://github.com/jhl-Det/RGBD-Video-SOD>